Question	Answer	Marks
1	**Defining the problem**	
	B is the independent variable and *v* is the dependent variable, or vary *B* measure *v*.	1
	Keep starting position of magnet constant/magnet always released from rest.	1
	Methods of data collection	
	Labelled diagram showing a magnet and the vertical copper tube supported.	1
	Method to ensure that copper tube is vertical, e.g. set square, spirit level, plumb line.	1
	Method to determine time at bottom of tube e.g. use of light gate(s)/motion sensor attached to timer/datalogger/computer or distance between two fixed marks at bottom of tube and stopwatch. Do not allow time over length of tube.	1
	Method to measure *B*, e.g. Hall probe.	1
	Method of analysis	
	Plot a graph of ln *v* against *B*.	1
	λ = – gradient	1
	$v_0 = e^{y\text{-intercept}}$	1

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Question	Answer	Marks
	Additional detail including safety considerations	**6**
	1. Keep mass of magnet constant. 2. Measurement of an appropriate length to determine v at bottom of tube, e.g. use ruler to measure distance between light gates/length of magnet/between two fixed marks. 3. $v = d / t$ for appropriate lengths (**not** length of tube) 4. Adjust Hall probe until maximum reading obtained/perpendicular to field/pole *or* Use Hall probe to take readings for both poles and average. 5. Method to calibrate Hall probe using a known field. 6. Safety precaution linked to falling magnets/use sand tray/cushion to soften fall. 7. Repeat experiment with magnets reversed and average *or* Repeat v (or t) for same B and average. 8. $\ln v = -\lambda B + \ln v_0$ 9. Relationship is valid if the graph is a straight line. 10. Method to vary B, e.g. re-magnetise in a coil.	

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